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TRIM BOARD CUTTING JIG ASSEMBLY AND METHODS OF USING THE SAME

Abstract

A reversible jig assembly for removing and replacing a portion of a trim board affixed to a substrate, comprising a first guide coupled to a first jig member; a second guide coupled to a second jig member, wherein the first guide and the second guide define a cutting channel for receiving a cutting blade, and wherein the first jig member and the second jig member are configured to define a first contact surface opposite a second contact surface; a first alignment rail extending a predetermined distance above and perpendicular to the first contact surface; and a second alignment rail extending a predetermined distance above and perpendicular to the second contact surface.

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Background/Summary

BACKGROUND

[0001] The described apparatus relates in general to cutting tools and accessories used in basic carpentry and more specifically to a trim cutting jig assembly for removing a predetermined portion of a trim board from a structure without having to remove the entire, full-length trim board from the structure.

[0002] Trim pieces and trim boards are typically installed on homes or other buildings to provide aesthetic appeal and to conceal intersections of walls and other surfaces. Exterior trim is often installed in the form of long, uniform boards that are mounted at various intersection points along the periphery of homes, adjacent to vinyl or other siding material. As with all exterior surfaces, trim boards often become damaged from weather or for other reasons and need to be removed or repaired. The use of conventional tools, such as hand saws or miter saws, to remove a portion of a trim board that is still attached to a structure typically results in damage to existing siding and to non-damaged trim. Imprecise cuts and injury to the user may also result from the use of such tools. Commercially available jigs and cutting devices typically require an entire trim board to be removed from a structure before making a desired cut in the trim board. If only a portion of a trim board is damaged, cost benefits and increased convenience could be realized by removing and replacing only the damaged portion without removing and replacing the entire trim board. Accordingly, there is an ongoing need for a trim cutting jig assembly that can be used to quickly and effectively remove only the portion of a trim board needing repair without having to completely remove the entire trim board from a structure.

SUMMARY

[0003] The following provides a summary of certain example implementations of the disclosed technology. This summary is not an extensive overview and is not intended to identify key or critical aspects or elements of the disclosed technology or to delineate its scope. However, it is to be understood that the use of indefinite articles in the language used to describe and claim the disclosed technology is not intended in any way to limit the described technology. Rather the use of “a” or “an” should be interpreted to mean “at least one” or “one or more”.

[0004] A first embodiment of the disclosed technology provides a jig assembly, comprising a first guide coupled to a first jig member; a second guide coupled to a second jig member, wherein the first guide and the second guide are configured to define a channel, and wherein the first jig member and the second jig member are configured to define a first contact surface opposite a second contact surface; and a first alignment rail extending a predetermined distance above and perpendicular to the first contact surface. Some implementations of the jig assembly further comprise a second alignment rail extending a predetermined distance above and perpendicular the second contact surface. In various implementations, the channel is a through-cut channel adapted to receive the blade of a cutting device. In various implementations, the first and second guides define a channel having a predetermined cutting angle, wherein the predetermined cutting angle is oriented upward at 30 degrees. In various implementations, the first alignment rail and the second alignment rail each include a plurality of recessed attachment devices that couple the first alignment rail and the second alignment rail to the first jig member and the second jig member respectively. Some implementations of the jig assembly further comprise a plurality of mounting holes formed in the first and second contact surfaces, wherein each mounting hole in the plurality of mounting holes is adapted to receive a mounting screw for attaching the jig assembly to a surface. In various implementations, the jig assembly is reversible such that the first contact surface and the second contact surface can each engage a portion of a trim board affixed to a substrate; and the predetermined cutting angle of the channel is maintained upward at 30 degrees. In various implementations, the jig assembly is operative to remove and replace the portion of a trim board while the trim board remains affixed to the substrate.

[0005] A second embodiment of the disclosed technology provides a reversible jig assembly for

removing and replacing a portion of a trim board affixed to a substrate, comprising a first guide coupled to a first jig member; a second guide coupled to a second jig member, wherein the first guide and the second guide are configured to define a through-cut channel that receives a blade, and wherein the first jig member and the second jig member are configured to define a first contact surface opposite a second contact surface; a first alignment rail extending a predetermined distance above and perpendicular to the first contact surface; and a second alignment rail extending a predetermined distance above and perpendicular to the second contact surface. In various implementations, the first and second guides define a through-cut channel having a predetermined cutting angle, wherein the predetermined cutting angle is oriented upward at 30 degrees relative to the trim board, and wherein the predetermined cutting angle of the through-cut channel is maintained upward at 30 degrees when the jig assembly is reversed. In various implementations, the first alignment rail and the second alignment rail each include a plurality of recessed attachment devices that couple the first alignment rail and the second alignment rail to the first jig member and the second jig member respectively.

[0006] A third embodiment of the disclosed technology provides a method for repairing a portion of a board, comprising mounting a reversible jig assembly at a predetermined location on a first board affixed to a structure, wherein the jig assembly includes a first guide coupled to a first jig member; and a second guide coupled to a second jig member, wherein the first guide and the second guide are configured to define a cutting channel; placing a blade in the cutting channel and making a cut through the first board; and removing a portion of the first board corresponding to the cut. Implementations of this method further comprise mounting the reversible jig assembly to a second board having dimensions that correspond to the removed section of the first board; placing the blade in the channel and making a cut through the second board, wherein the angle of the cut in the second board corresponds to the angle of the cut made in the first board; and replacing the removed portion of the first board with the cut portion of the second board and affixing the second board to the structure. In various implementations, the first and second guides define a channel having a predetermined cutting angle, wherein the predetermined cutting angle is oriented upward at 30 degrees relative to the first board, and wherein the predetermined cutting angle of the channel is maintained upward at 30 degrees when the jig assembly is reversed. In various implementations, the first jig member and the second jig member define a first contact surface opposite a second contact surface. In various implementations, the reversible jig assembly further comprises a first alignment rail coupled to the first jig member and the second jig member, the first alignment rail extending a predetermined distance above and perpendicular to the first contact surface; and a second alignment rail coupled to the first jig member and the second jig member, the second alignment rail extending a predetermined distance above and perpendicular to the second contact surface. In various implementations, the first alignment rail and the second alignment rail each include a plurality of recessed attachment devices that couple the first alignment rail and the second alignment rail to the first jig member and the second jig member respectively.

[0007] It should be appreciated that all combinations of the foregoing concepts and additional concepts discussed in greater detail below (provided such concepts are not mutually inconsistent) are contemplated as being part of the technology disclosed herein and may be implemented to achieve the benefits as described herein. Additional features and aspects of the disclosed system, devices, and methods will become apparent to those of ordinary skill in the art upon reading and understanding the following detailed description of the example implementations. As will be appreciated by the skilled artisan, further implementations are possible without departing from the scope and spirit of what is disclosed herein. Accordingly, the drawings and associated descriptions are to be regarded as illustrative and not restrictive in nature.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] The accompanying drawings, which are incorporated into and form a part of the specification, schematically illustrate one or more example implementations of the disclosed technology and, together with the general description given above and detailed description given below, serve to explain the principles of the disclosed subject matter, and wherein:

[0009] FIG. 1 is a first perspective view of an example implementation of the disclosed trim cutting jig assembly;

[0010] FIG. 2 is an alternate perspective view of the trim cutting jig assembly of FIG. 1;

[0011] FIG. 3 is a cross-sectional, top down view of the trim cutting jig assembly of FIG. 1;

[0012] FIG. 4 is a cross-sectional, side view of the trim cutting jig assembly of FIG. 1;

[0013] FIG. 5 is a front view of the trim cutting jig assembly of FIG. 1;

[0014] FIG. 6A depicts the trim cutting jig assembly of FIG. 1 mounted to an exemplary trim board on an existing structure or substrate; and

[0015] FIG. 6B depicts the trim cutting jig assembly of FIG. 2 mounted to an exemplary trim board on an existing structure or substrate.

DETAILED DESCRIPTION

[0016] Example implementations are now described with reference to the Figures. Reference numerals are used throughout the detailed description to refer to the various elements and structures. Although the following detailed description contains many specifics for the purposes of illustration, a person of ordinary skill in the art will appreciate that many variations and alterations to the following details are within the scope of the disclosed technology. Accordingly, the following implementations are set forth without any loss of generality to, and without imposing limitations upon, the claimed subject matter.

[0017] The examples discussed herein are examples only and are provided to assist in the explanation of the apparatuses, devices, systems, and methods described herein. None of the features or components shown in the drawings or discussed below should be taken as required for any specific implementation of any of these the apparatuses, devices, systems or methods unless specifically designated as such. For ease of reading and clarity, certain components, modules, or methods may be described solely in connection with a specific Figure. Any failure to specifically describe a combination or sub-combination of components should not be understood as an indication that any combination or sub-combination is not possible. Also, for any methods described, regardless of whether the method is described in conjunction with a flow diagram, it should be understood that unless otherwise specified or required by context, any explicit or implicit ordering of steps performed in the execution of a method does not imply that those steps must be performed in the order presented but instead may be performed in a different order or in parallel.

[0018] Example implementations of the disclosed systems, devices, and methods provide a trim cutting device or trim cutting jig assembly for removing a portion of a trim board or other board affixed to a structure or substrate, such as a home or building, without having to completely remove and replace the full trim board. The disclosed trim cutting device is reversible to maintain an upward, 30 degree cutting angle relative to the trim board/substrate, thereby channeling water away from the cut and mitigating water intrusion, while also preventing damage to existing siding and trim during cutting. As discussed herein: (i) the trim cutting device can be mounted to an existing board already affixed to a structure; (ii) a multi-tool or similar tool having a blade can engage the trim cutting device at an upward cutting angle and make a consistent, uniform cut along the board at a predetermined location; and (iii) when the trim cutting device is unmounted, the cut portion of the board can be removed. The disclosed trim cutting device can thereafter be attached to a new, independent replacement board having the same dimensions as the removed board portion to obtain a corresponding cut. The replacement board can then be nailed to the existing board affixed to a structure, and all remaining holes can be caulked.

[0019] FIG. 1 provides a first perspective view of an example implementation of the disclosed trim cutting jig assembly; FIG. 2 provides a second perspective view of the trim cutting jig assembly of FIG. 1; FIG. 3 is a cross-sectional, top down view of the trim cutting jig assembly of FIG. 1; FIG. 4 is a cross-section, side view of the trim cutting jig assembly of FIG. 1; and FIG. 5 is a front view of the trim cutting jig assembly of FIG. 1.

[0020] With specific reference to FIGS. 1-5, trim cutting jig assembly **10** includes first jig member **100** and second jig member **200**. First jig member **100** includes first contact surface **105**, second contact surface **110**, outward facing edge **115**, inward facing edge **120**, and side edges **125**, **130**. Second jig member **200** includes first contact surface **205**, second contact surface **210**, outward facing edge **215**, inward facing edge **220**, and side edges **225**, **230**. A plurality of apertures **20** having inserts **25** are formed in side edges **125**, **225** and **130**, **230** for receiving screws **30** or other attachment devices. Screws **30** attach first alignment rail **300** to side edges **125**, **225** and second alignment rail **400** to side edges **130**, **230**. First alignment rail **300** and second alignment rail **400** affix first jig member **100** and second jig member **200** at a predetermined distance from one another to define through-cut channel **500** having a predetermined angle, which in this example is 30 degrees. In the present exemplary embodiment, screws **30** are recessed within first alignment rail **300** and second alignment rail **400**. When fully assembled, (i) first contact surfaces **105**, **205** are substantially co-planar; and (ii) second contact surfaces **110**, **210** are substantially co-planar. First alignment rail **300** includes edge **310** that extends in a first direction at a predetermined distance above and perpendicular to first contact surfaces **105**, **205**. Second alignment rail **400** includes edge **410** that extends in a second direction at a predetermined distance above and perpendicular to second contact surfaces **110**, **210**.

[0021] With specific reference to FIGS. 3-4, a plurality of holes **40** having inserts **45** are formed in inward facing edges **120**, **220** for receiving screws **50** or other attachment devices that affix first guide **150** to first jig member **100** and second guide **250** to second jig member **200**. First guide **150** and second guide **250** extend the full length of inward facing edge **120** and inward facing edge **220**, respectively, and are configured to guide a blade of a multi-tool, or similar cutting device, within through-cut channel **500** at a predetermined angle along cut-line **85**.

[0022] With reference to FIGS. 1-3 and 5, a plurality of storage holes **60** are formed in outward facing edges **115**, **215** that hold screws **70** or other attachment devices during nonuse of the trim cutting jig assembly **10**. A plurality of mounting holes **80** are coaxially formed in first contact surfaces **105**, **205** and second contact surfaces **110**, **210** for receiving screws **70** when mounting the fully assembled trim cutting jig assembly **10** (see FIGS. 1-2, receiving screws **70** in dashed/broken lines).

[0023] FIG. 6A depicts the disclosed trim cutting jig assembly **10** mounted to an exemplary trim board on an existing structure or substrate; and FIG. 6B depicts the disclosed trim cutting jig assembly **10** reversibly mounted to another exemplary trim board on an existing structure or substrate. In FIG. 6A, screws **70** are fastened within mounting holes **80** to trim board **90**, thereby affixing trim cutting jig assembly **10** to a portion of trim board **90** to be removed. In this non-limiting example, trim cutting jig assembly **10** is fastened to trim board **90** such that (i) second contact surfaces **110**, **210** are flush with the surface of trim board **90**; and (ii) through-cut channel **500** is configured at an upward 30 degree cutting angle relative to trim board **90**. Edge **310** of first alignment rail **300** extends away from trim board **90** to prevent contact with adjacent trim board **95** during cutting; and recessed screws **30** in second alignment rail **400** permit second alignment rail **400** to fit between trim board **90** and siding **92** without damaging siding **92**. Once trim cutting jig assembly **10** is securely mounted to trim board **90**, through-cut channel **500** receives the blade of a cutting device to form a consistent cut along the portion of trim board **90** to be removed. Screws **70** are then removed from mounting holes **80** so that the portion of trim board **90** can be completely removed.

[0024] In FIG. 6B, trim cutting jig assembly **10** is reversed and attached to trim board **95** such that:

(i) first contact surfaces **105**, **205** are flush with the surface of trim board **95**; (ii) through-cut channel **500** is maintained at an upward 30 degree cutting angle relative to trim board **95**; (iii) edge **410** of second alignment rail **400** extends away from trim board **95** to prevent contact with adjacent trim board **90** during cutting; (iv) recessed screws **30** in first alignment rail **300** permit first alignment rail **300** to fit between trim board **95** and siding **97** without damaging siding **97**. The reversibility of trim cutting jig assembly **10** creates a consistent 30 degree cut upwards towards the substrate to channel water away from the cut and mitigate water intrusion.

[0025] In non-limiting examples, the inserts used with the disclosed trim cutting jig assembly are 18-8 stainless steel helical inserts, 8-32 right-hand thread, and 0.164" long; and the screws used with the disclosed trim cutting jig assembly are zinc-plated alloy steel hex drive flat head screw having 8-32 thread size, and 0.25" long.

[0026] All literature and similar material cited in this application, including, but not limited to, patents, patent applications, articles, books, treatises, and web pages, regardless of the format of such literature and similar materials, are expressly incorporated by reference in their entirety. Should one or more of the incorporated references and similar materials differ from or contradict this application, including but not limited to defined terms, term usage, described techniques, or the like, this application controls.

[0027] As previously stated and as used herein, the singular forms "a," "an," and "the," refer to both the singular as well as plural, unless the context clearly indicates otherwise. The term "comprising" as used herein is synonymous with "including," "containing," or "characterized by," and is inclusive or open-ended and does not exclude additional, unrecited elements or method steps. Although many methods and materials similar or equivalent to those described herein can be used, particular suitable methods and materials are described herein. Unless context indicates otherwise, the recitations of numerical ranges by endpoints include all numbers subsumed within that range. Furthermore, references to "one implementation" are not intended to be interpreted as excluding the existence of additional implementations that also incorporate the recited features. Moreover, unless explicitly stated to the contrary, implementations "comprising" or "having" an element or a plurality of elements having a particular property may include additional elements whether or not they have that property.

[0028] The terms "substantially" and "about", if or when used throughout this specification describe and account for small fluctuations, such as due to variations in processing. For example, these terms can refer to less than or equal to $\pm 5\%$, such as less than or equal to $\pm 2\%$, such as less than or equal to $\pm 1\%$, such as less than or equal to $\pm 0.5\%$, such as less than or equal to $\pm 0.2\%$, such as less than or equal to $\pm 0.1\%$, such as less than or equal to $\pm 0.05\%$, and/or 0%.

[0029] Underlined and/or italicized headings and subheadings are used for convenience only, do not limit the disclosed subject matter, and are not referred to in connection with the interpretation of the description of the disclosed subject matter. All structural and functional equivalents to the elements of the various implementations described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and intended to be encompassed by the disclosed subject matter. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the above description.

[0030] There may be many alternate ways to implement the disclosed technology. Various functions and elements described herein may be partitioned differently from those shown without departing from the scope of the disclosed technology. Generic principles defined herein may be applied to other implementations. Different numbers of a given module or unit may be employed, a different type or types of a given module or unit may be employed, a given module or unit may be added, or a given module or unit may be omitted.

[0031] Regarding this disclosure, the term "a plurality of" refers to two or more than two. Unless otherwise clearly defined, orientation or positional relations indicated by terms such as "upper" and

“lower” are based on the orientation or positional relations as shown in the FIGS., only for facilitating description of the disclosed technology and simplifying the description, rather than indicating or implying that the referred devices or elements must be in a particular orientation or constructed or operated in the particular orientation, and therefore they should not be construed as limiting the disclosed technology. The terms “connected”, “mounted”, “fixed”, etc. should be understood in a broad sense. For example, “connected” may be a fixed connection, a detachable connection, or an integral connection; a direct connection, or an indirect connection through an intermediate medium. For an ordinary skilled in the art, the specific meaning of the above terms in the disclosed technology may be understood according to specific circumstances.

[0032] It should be appreciated that all combinations of the foregoing concepts and additional concepts discussed in greater detail herein (provided such concepts are not mutually inconsistent) are contemplated as being part of the disclosed technology. In particular, all combinations of claimed subject matter appearing at the end of this disclosure are contemplated as being part of the technology disclosed herein. While the disclosed technology has been illustrated by the description of example implementations, and while the example implementations have been described in certain detail, there is no intention to restrict or in any way limit the scope of the appended claims to such detail. Additional advantages and modifications will readily appear to those skilled in the art. Therefore, the disclosed technology in its broader aspects is not limited to any of the specific details, representative devices and methods, and/or illustrative examples shown and described. Accordingly, departures may be made from such details without departing from the spirit or scope of the general inventive concept.

Claims

1. A jig assembly, comprising: (a) a first guide coupled to a first jig member; (b) a second guide coupled to a second jig member, wherein the first guide and the second guide are configured to define a channel, and wherein the first jig member and the second jig member are configured to define a first contact surface opposite a second contact surface; and (c) a first alignment rail extending a predetermined distance above and perpendicular to the first contact surface.
2. The jig assembly of claim 1, further comprising a second alignment rail extending a predetermined distance above and perpendicular to the second contact surface.
3. The jig assembly of claim 1, wherein the channel is a through-cut channel adapted to receive the blade of a cutting device.
4. The jig assembly of claim 1, wherein the first and second guides define a channel having a predetermined cutting angle.
5. The jig assembly of claim 4, wherein the predetermined cutting angle is oriented upward at 30 degrees.
6. The jig assembly of claim 2, wherein the first alignment rail and the second alignment rail each include a plurality of recessed attachment devices that couple the first alignment rail and the second alignment rail to the first jig member and the second jig member respectively.
7. The jig assembly of claim 1, further comprising a plurality of mounting holes formed in the first and second contact surfaces, wherein each mounting hole in the plurality of mounting holes is adapted to receive a mounting screw for attaching the jig assembly to a surface.
8. The jig assembly of claim 4, wherein the jig assembly is reversible such that: (a) the first contact surface and the second contact surface can each engage a portion of a trim board affixed to a substrate; and (b) the predetermined cutting angle of the channel is maintained upward at 30 degrees.
9. The jig assembly of claim 8, wherein the jig assembly is operative to remove and replace the portion of a trim board while the trim board remains affixed to the substrate.
10. A reversible jig assembly for removing and replacing a portion of a trim board affixed to a

substrate, comprising: (a) a first guide coupled to a first jig member; (b) a second guide coupled to a second jig member, wherein the first guide and the second guide are configured to define a through-cut channel that receives a blade, and wherein the first jig member and the second jig member are configured to define a first contact surface opposite a second contact surface; (c) a first alignment rail extending a predetermined distance above and perpendicular to the first contact surface; and (d) a second alignment rail extending a predetermined distance above and perpendicular the second contact surface.

11. The jig assembly of claim 10, wherein the first and second guides define a through-cut channel having a predetermined cutting angle.

12. The jig assembly of claim 11, wherein the predetermined cutting angle is oriented upward at 30 degrees relative to the trim board, and wherein the predetermined cutting angle of the through-cut channel is maintained upward at 30 degrees when the jig assembly is reversed.

13. The jig assembly of claim 10, wherein the first alignment rail and the second alignment rail each include a plurality of recessed attachment devices that couple the first alignment rail and the second alignment rail to the first jig member and the second jig member respectively.

14. A method for repairing a portion of a board, comprising: (a) mounting a reversible jig assembly at a predetermined location on a first board affixed to a structure, wherein the jig assembly includes: (i) a first guide coupled to a first jig member; and (ii) a second guide coupled to a second jig member, wherein the first guide and the second guide are configured to define a cutting channel; (b) placing a blade in the cutting channel and making a cut through the first board; and (c) removing a portion of the first board corresponding to the cut.

15. The method of claim 10, further comprising: (a) mounting the reversible jig assembly to a second board having dimensions that correspond to the removed section of the first board; (b) placing the blade in the channel and making a cut through the second board, wherein the angle of the cut in the second board corresponds to the angle of the cut made in the first board; and (c) replacing the removed portion of the first board with the cut portion of the second board and affixing the second board to the structure.

16. The method of claim 14, wherein the first and second guides define a channel having a predetermined cutting angle.

17. The method of claim 16, wherein the predetermined cutting angle is oriented upward at 30 degrees relative to the first board, and wherein the predetermined cutting angle of the channel is maintained upward at 30 degrees when the jig assembly is reversed.

18. The method of claim 14, wherein the first jig member and the second jig member define a first contact surface opposite a second contact surface.

19. The method of claim 14, wherein the reversible jig assembly further comprises: (a) a first alignment rail coupled to the first jig member and the second jig member, the first alignment rail extending a predetermined distance above and perpendicular to the first contact surface; and (d) a second alignment rail coupled to the first jig member and the second jig member, the second alignment rail extending a predetermined distance above and perpendicular to the second contact surface.

20. The method of claim 15, wherein the first alignment rail and the second alignment rail each include a plurality of recessed attachment devices that couple the first alignment rail and the second alignment rail to the first jig member and the second jig member respectively.
